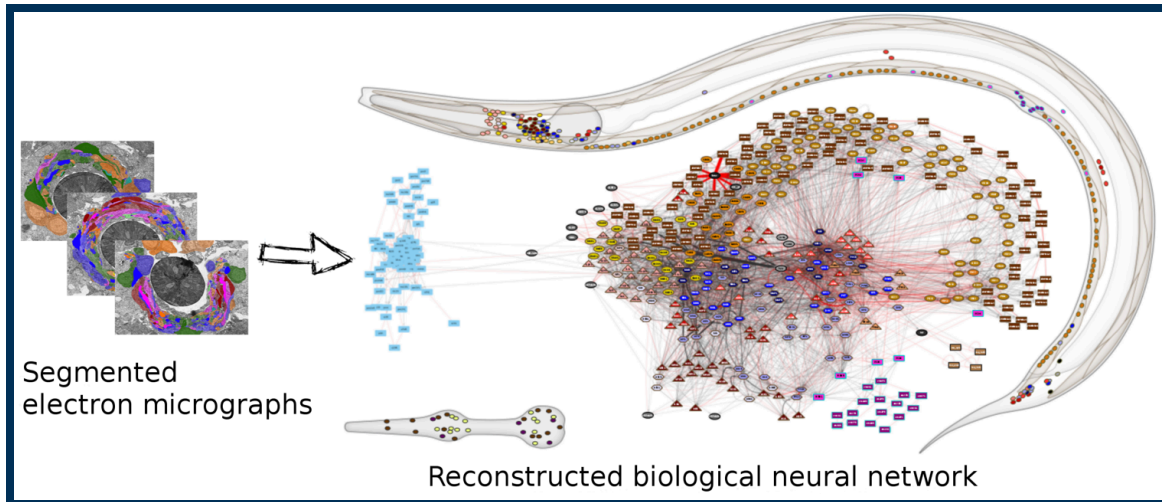


L1: Example: The Brain of the C.elegans Worm



(Image Source: wormwiring.org → [\(https://wormwiring.org/\)](https://wormwiring.org/))

To understand the relationship between a complex system and its network representation, let's focus on a microscopic worm called C.elegans.

This amazing organism, which is *about 1mm in length*, has roughly only 300 neurons in its neural system. Still, it can move in different ways, react to touch, mate, sense chemical odors, and respond to food versus toxins, etc.

Each dot represents a neuron, and the location of every neuron at the worm's body is shown at the top right. The connections between neurons, at the level of individual synapses, have been mapped using electron micrographs of tiny slices of the worm's body.

The network on the right in the image shows each neuron as a node and each connection between two neurons as an edge between the corresponding two nodes. Do not worry about the different colors for now – we will discuss this network again later in the course. The important point, for now, is that network science maps this highly complex living system into a graph -- an abstraction that we can analyze mathematically and computationally to ask a number of important questions about the organization of this neural system.

Note that this mapping from an actual system to a graph representation is only a model, and so it discards some information that the modeler views as non-essential. For instance, the network representation does not show in this example if a neuron is excitatory or inhibitory -- or whether these connections remain the same during the worm's lifetime.

So it is always important to ask: *does the network representation of a given system provide sufficient detail to answer the questions we are really interested in about that system?*